

What Each Task Does — and Why It Matters

Stage 0 — Prerequisites

Overall goal: get every basic tool and decision in place before real work starts, so Stage 1 isn't lost to setup delays.

0.1 Cloud Setup

What it is: Creating the GCP project, activating the free voucher, installing and authenticating the gcloud CLI, and deciding which GCP services the project needs (BigQuery, Cloud Run, Vertex AI).

Why it matters: Nothing in Pillar 1 can start without a working GCP project to build in — this is the empty foundation everything else gets built on top of.

0.2 Environment Setup

What it is: Committing .gitignore and requirements.txt, documenting the Python 3.12+ virtual environment setup, and confirming everyone's IDE has the right extensions.

Why it matters: Keeps everyone's local setup consistent, so "it works on my machine" doesn't become a recurring problem once real code starts getting written.

0.3 Architecture Sketch v1

What it is: Creating the shared Excalidraw canvas and drawing the first rough version of the whole system — logical flow plus GCP infrastructure.

Why it matters: Required directly by the brief. It's also the one artifact that lets anyone — teammate, mentor, or eventually a client — understand the whole project from a single picture before reading any code.

0.4 Dataset & KPI Draft

What it is: Confirming and downloading the Global Electronics Retailer dataset, and drafting the initial KPI list.

Why it matters: You can't design a schema or write a single query without first knowing what data you actually have and what business questions it needs to answer.

Stage 1 — Contracts & Skeleton

Overall goal: agree on shared interfaces so all four pillars can be built in parallel without anyone waiting on anyone else.

1.1 Write schema.md

What it is: Documenting every table and column name, data type, and relationship in the dataset, plus a business-terms-to-SQL mapping.

Why it matters: This is the single reference the LLM, the data pipeline, and anyone hand-writing SQL all rely on — without it, everyone would be guessing at column names independently and getting different answers.

1.2 Write api.md

What it is: Documenting the exact request and response shape for the /ask endpoint, including the normal, clarification-needed, and error cases.

Why it matters: This is what lets the frontend and backend be built by different people at the same time — each builds against the agreed shape rather than waiting for the other's real code to exist.

1.3 Repo Skeleton

What it is: Creating the /src subfolders (data, llm, backend, frontend), /mock_data, and /docs folders in GitHub.

Why it matters: Gives every pillar a clear, agreed home for its code from day one — no confusion later about where anyone's work is supposed to go.

Stage 2 — Core Build (Four Pillars)

Overall goal: build the actual system, one pillar at a time, whole team on each.

Pillar 1 — Cloud & Data Platform

Goal: get TGS's sales data into a real, queryable database. Nothing else in the project can work without this.

1.1 Retail Data Cleaning & ETL

What it is: Taking the raw CSV files and turning them into clean, consistent tables — fixing formatting issues, handling missing values, making sure dates and numbers are in the right type.

Why it matters: Garbage in, garbage out. If the data is messy, every KPI and every SQL query built on top of it will be wrong, no matter how good the AI is.

1.2 BigQuery Data Warehouse Setup

What it is: Creating the actual database tables in BigQuery, deciding how they're organized (partitioning, indexing) for fast queries.

Why it matters: This is the database the whole system queries. Good structure here means fast, cheap queries later; bad structure means slow, expensive ones as the project scales.

1.3 IAM & Security Configuration

What it is: Setting up who/what is allowed to access the database, using least-privilege principles (nobody gets more access than they need).

Why it matters: Good practice for any real system — shows you're thinking about security, not just "does it work," which matters in a professional context.

1.4 Docker & Cloud Run Deployment

What it is: Packaging the app into containers so it can run on Google Cloud Run instead of just on someone's laptop.

Why it matters: This is what turns "a demo on my machine" into "a live link anyone can open" — required for your actual final presentation.

Pillar 2 — LLM & Text-to-SQL

Goal: turn a plain-English question into a correct, safe SQL query. This is the heart of "Talk-to-my-data" — literally the thing the brief is asking you to build.

2.1 Dynamic Schema Injection

What it is: Automatically feeding the LLM a description of your database tables/columns every time it's asked a question.

Why it matters: This is word-for-word what the brief requires — "enhance the user's question with knowledge about the database tables." Without this, the AI is guessing at table names; with it, the AI knows exactly what exists.

2.2 Few-Shot Prompting & Domain Lexicon

What it is: Giving the LLM a handful of example questions and their correct SQL, so it has a pattern to follow, plus a glossary mapping business words ("revenue," "returns") to precise SQL logic.

Why it matters: Makes the AI dramatically more consistent and accurate — the difference between it guessing what "revenue" means each time vs. always calculating it the same, correct way.

2.3 Self-Healing Query Feedback Loop

What it is: If the generated SQL has an error, automatically show that error back to the LLM and ask it to fix itself, rather than just failing.

Why it matters: Real questions are messy and the first attempt doesn't always work — this makes the system resilient instead of brittle, which is what separates a toy demo from something genuinely usable.

2.4 Ambiguity Resolution & Clarification

What it is: Detecting when a question could mean more than one thing (e.g. "sales" = revenue or units?) and asking the user to clarify instead of guessing.

Why it matters: A wrong confident answer is worse than an honest question — this is what makes stakeholders actually trust the tool with real business decisions.

Pillar 3 — Backend & Analytics Engine

Goal: be the safe, reliable middle layer that connects everything — takes the question in, coordinates the AI and the database, and sends a proper answer back out.

3.1 FastAPI Core Service Layer

What it is: The actual server that receives questions from the UI, calls the LLM, calls BigQuery, and returns results — the /ask endpoint from api.md.

Why it matters: This is the glue. Without it, the frontend and the AI/database logic have no way to talk to each other.

3.2 SQL Validation & Safety Guardrails

What it is: Checking every piece of generated SQL before running it, to make sure it's a harmless SELECT and never something that could change or delete data.

Why it matters: You're letting an AI write and run database queries automatically — this is the non-negotiable safety net that makes that acceptable to do at all.

3.3 Retail KPI Computation Engine

What it is: The actual logic that calculates business metrics — revenue, average order value, category share — consistently, every time.

Why it matters: Directly required by the brief — "compute at least 2 KPIs." This is also literally what TGS's head of sales cares about; it's the business value of the whole project.

3.4 Multi-Modal Response Formatter

What it is: Taking the raw rows that come back from BigQuery and turning them into three things at once: a plain-English summary, a data table, and chart-ready data.

Why it matters: Raw SQL output is unreadable to a business stakeholder — this is what makes the answer actually useful and understandable to a non-technical person.

Pillar 4 — Frontend UI, Benchmark & Delivery

Goal: make the system usable by an actual human, and prove it actually works well.

4.1 Streamlit Conversational Interface

What it is: The chat window itself — where someone types a question and sees the answer.

Why it matters: This is the only part of the whole project a TGS stakeholder ever actually sees. Everything else exists to feed this screen.

4.2 Dynamic Data Visualizations

What it is: Turning the chart data from the formatter into actual on-screen graphs (line charts for trends, bar charts for comparisons).

Why it matters: A number in a table is harder to grasp at a glance than a chart — this makes insights land faster for a business audience.

4.3 Golden Benchmark Evaluation Suite

What it is: *A set of test questions with known correct answers, used to measure how often the system actually gets it right.*

Why it matters: This is what turns "we think it works" into "we measured that it's 90% accurate" — the difference between a claim and proof.

4.4 Excalidraw Architecture & Client Deck

What it is: *The living system diagram, and the slides that pitch the project to Mark and Quinten.*

Why it matters: Directly required by the brief. Also genuinely the thing that determines whether people understand and are convinced by everything else you built.

Stage 3 — Integration

Overall goal: make the four independently-built pillars actually work together as one system.

Full Pipeline Integration

What it is: *Wiring the real data pipeline, real LLM logic, real backend, and real frontend together, replacing any remaining mocks with the genuine components.*

Why it matters: Each pillar working alone proves the parts are individually solid — this stage proves the whole system actually works end-to-end, which is what you're ultimately graded on and what stakeholders will actually use.

Stage 4 — Dry Run Prep

Overall goal: get honest feedback from Mark and Quinten while there's still a week to act on it.

Dry Run Deck + Rehearsal

What it is: *Building a simple slide deck covering the chosen problem, architecture, first results, and a working (if unpolished) live demo, then rehearsing it as a team before presenting.*

Why it matters: The whole point of a dry run is catching problems early — in the pitch, the demo, or the plan — while there's still time to fix them, rather than discovering issues for the first time at the final presentation.

Stage 5 — Final Polish & Deployment

Overall goal: deliver a fully deployed, working system with a convincing business pitch.

Cloud Deployment + Final Deck

What it is: *Deploying the finished app fully to Cloud Run, polishing the final presentation deck (business case, architecture, results vs. KPIs, rollout advice), and rehearsing the live demo.*

Why it matters: This is the version that gets judged, gets reused as a portfolio piece in interviews, and is the last impression your team leaves with Artefact — worth the extra polish.

The Pipeline in One Sentence

A user asks a question in plain English → the AI, knowing your database structure, turns it into safe SQL → the backend validates and runs it against real data in BigQuery → the results come back formatted as a summary, table, and chart → all shown in a chat interface anyone at TGS could actually use.